

An Analysis of Arbitrage Opportunities on S&P 500 ETFs

1 Introduction

1.1 What are ETFs?

ETF stands for exchange traded fund. ETFs are a type of investment fund that hold many underlying securities, such as stocks, bonds, or commodities, offering an investor exposure to a certain portfolio. ETFs can track market indices like the S&P-500, for example. Also, notably, ETFs are tradeable themselves intraday on exchanges (Chen, 2026).

1.2 What is ETF arbitrage?

ETF arbitrage is the process of exploiting price differences between an ETF and its underlying basket of constituent securities. These price differences arise because of short-term supply and demand dynamics of the ETF and/or basket of securities in the market. Another form of ETF arbitrage is a pairs trading strategy between two related ETFs, for example if they track the same index. In this strategy, the price difference between the two ETFs is exploited (Zucchi, 2025).

1.3 Key Terms

Net Asset Value (NAV): NAV represents the fair value of an ETF. It is calculated by subtracting total liabilities from total assets plus cash and dividing by shares outstanding.

Unit Investment Trust (UIT): A UIT is a form of management investment company that represents a fixed portfolio with a fixed number of units (shares) and a set expiration date. Since the number of units is fixed, shares of UITs trade in the secondary market and are influenced by supply and demand.

Open-Ended Fund: An open-ended fund is another form of management investment company with additional capabilities to reinvest dividends, for example. In the context of ETFs, open-ended funds are exchange traded.

1.4. S&P-500 ETFs

The following are popular S&P-500 ETFs considered for this study.

State Street SPDR S&P 500 ETF Trust (SPY)	First U.S.-listed exchange-traded fund (ETF); UIT
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Vanguard S&P 500 ETF (VOO)	Open-ended ETF
iShares Core S&P 500 ETF (IVV)	Open-ended ETF
Direxion Daily S&P 500 Bull 3X ETF (SPXL)	Leveraged ETF designed to enhance returns in a bull market

2. Project Overview

For this project, we analyze arbitrage opportunities in S&P-500 ETFs. Given that this opportunity is largely exploited by high-frequency trading firms, analysis of these opportunities requires tick-level price and quantity data for the ETFs and underlying basket stocks. We detail the many data collection and cleaning processes and challenges that we encountered in Section 2. In Section 3, we analyze arbitrage opportunities for S&P-500 ETFs. We consider opportunities during normal trading periods and during market news events that lead to higher volatility across different S&P-500 ETFs. We also present an analysis of pairs trading between S&P-500 ETFs themselves and a lead-lag relationship analysis. In Section 4, we discuss the results of our analysis and the implications for practitioners.

2. Data

2.1 Data Collection

For price and quantity data of the ETFs and underlying basket stocks, we utilized Wharton Reserve Data Services to access the NYSE Trades and Quotes (TAQ) Dataset, which provides access to the intraday national best bid and offer (NBBO) of 10,000 stock issues listed on 16 major American exchanges, including the New York Stock Exchange (NYSE), American Stock Exchange (AMEX), and Nasdaq. Specifically, we used the Trade & Quote Daily Product, which is updated daily and has timestamps with microsecond granularity as of April 2015 (Wharton Research Data Services, n.d.). For daily fund flows and constituent data, we utilize the ETF Fund Flows and ETF Constituents datasets from ETF Global accessed on *massive.com* (ETF Global, 2026). This data is necessary to produce a fair intraday net asset value of the ETF from underlying basket stock prices, which will be discussed in Section 2.3.

2.2 Data Aggregation

Since the intraday price and quantity data was collected locally, we faced a number of

challenges in managing the data files. In order to make the collection process efficient, we queried the NBBO of the desired ETF and all underlying basket constituents in 15 second increments. The resulting data files contained approximately 50,000 data points for each category (bid price, ask price, bid quantity, ask quantity), sometimes far more during market news events considered like FOMC meetings. Across an entire trading day, this process results in 78 million data points across all categories. Needless to say, this resulted in a computational challenge. Even using parallel execution across different products and trading days, it often took more than 12 hours to aggregate a singular day's worth of data. In practice, I believe this could be mitigated in a number of ways. The two bottlenecks in our data aggregation program was the TAQ query through WRDS on the underlying constituent stock data and creating a pivot table to aggregate the data for each ticker. We are obviously restricted by the query to the WRDS database, and in a real trading environment, we believe the access to clean historical data would be more readily available. However, practitioners may disagree as I am purely hypothesizing. In regards to building the pivot table to aggregate the data by underlying basket ticker, there are alternatives to the basic Python implementation that we used through the *pandas* library (The pandas development team, 2026). Some alternatives to explore for future work are the *polars* Python library which claims to provide faster computation on large datasets or, from a programming perspective, reordering the order of computation, avoiding the pivot table all together.

2.3 Computing a “Fair Value”

In order to compute a fair value for the underlying basket, which in this case would also be a proxy for the intraday net asset value (NAV) of the ETF, it requires a fair value estimate of each underlying constituent stock. Rather than using the last trade price, we use the weighted mid-price, as discussed in class, which uses the feature of imbalance. The formula for the weighted mid-price is

$$W_{mid} = B_{price}(1 - I) + A_{price}I$$

where I is imbalance computed as

$$I = \frac{B_{quantity}}{B_{quantity} + A_{quantity}}$$

and B is bid and A is ask. The weighted mid-price exhibits many of the properties of microprices, including stability and ease of computing; however, it is not necessarily a martingale, and it has been shown that at large spreads, which occurs in some illiquid products, the weighted mid-price converges

to the mid-price. Perhaps the micro-price would have been a better estimate, particularly for more illiquid tickers in the underlying basket constituents, but we chose the weighted-mid price for familiarity and because price movements in the underlying basket are largely driven by price movements in more heavily weighted stocks by market capitalization, which are typically more liquid (Netze & Palige, 2017). Using quotes from the NYSE TAQ dataset introduced in Section 2.1, we are able to compute this value for both the ETF and the underlying basket constituent stock. More generally, we consider this price the fair value of the stock or ETF.

Once a fair value estimate is computed for each underlying basket constituent stock for the given ETF, we are able to compute a fair value for the underlying basket. The net asset value of an ETF is computed by the following formula

$$NAV = \frac{Total\ Assets + Cash - Total\ Liabilities}{Shares\ Outstanding}$$

which for the purposes of an S&P-500 ETF results in

$$NAV = \frac{\Sigma\ Share\ price\ (\$) * Shares\ held + Cash\ (\$)}{Shares\ Outstanding}$$

where $\Sigma\ Share\ price\ (\$) * Shares\ held$ corresponds to the sum of the share price, for which we use the weighted mid-price to account for the microsecond timescale granularity, times the quantity of shares held by the ETF, which is published by the ETF provider daily. Using the ETF Fund Flows and ETF Constituents datasets introduced in Section 2.1, we are able to compute this value using the weighted mid-prices discussed previously. Unfortunately, while this value captured the general price movements of the ETF, it exhibited a certain drift over the trading day. We are unsure as to what caused this phenomenon. However, to account for it, we computed a scale factor that is applied periodically throughout the trading day to ensure that the values are aligned. We believe that this is justified because we are focusing this project on low-latency arbitrage opportunities, so we can assume that these larger price differences are a result of incomplete datasets or are addressed by authorized participants through creation redemption, as discussed in Section 4.2. Additionally, we collect the bid and ask prices for the ETF itself and compute the aggregated bid and ask prices for the underlying constituent basket to understand the transaction costs involved with buying or selling the underlying basket and ETF, respectively. Figure 1 shows a plot of these prices for SPY on July 15, 2024. As illustrated, the fair value price of the underlying basket closely tracks the fair value of the ETF, and the basket spread is much wider than the spread of the individual ETF because of the aggregated

transaction costs across ~500 constituent tickers. We perform the described method of data collection and aggregation for two standard S&P-500 ETFs and one leveraged S&P-500 ETF across various trading days and surrounding certain market events.

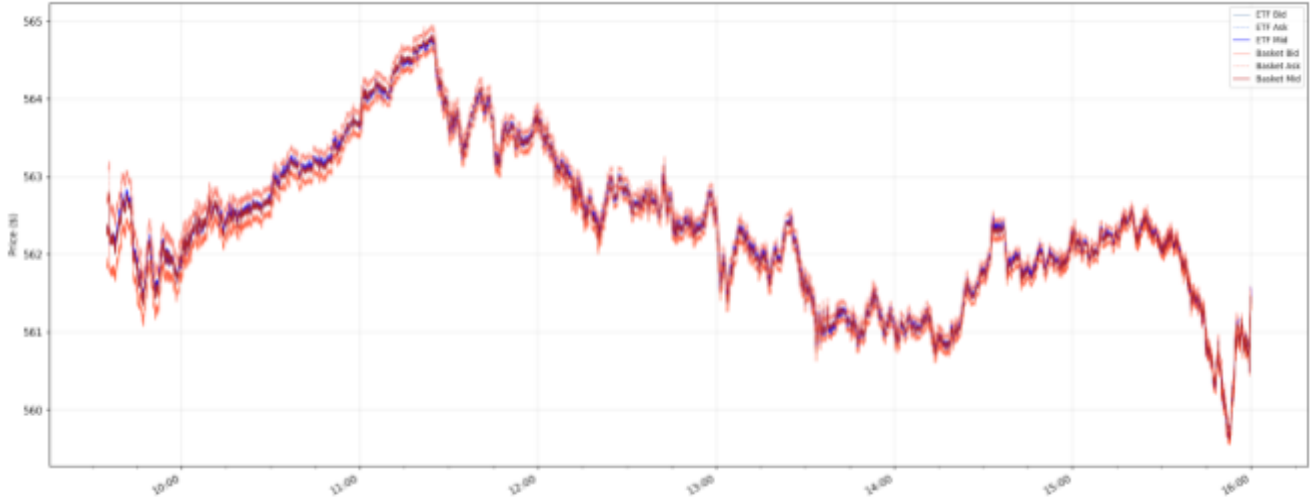


Figure 1. ETF (SPY) and underlying basket bid, ask, and mid prices for July 15, 2024.

3. Analysis

To analyze the arbitrage opportunities for S&P-500 ETFs, we consider the length and magnitude of price discrepancies between ETFs and between ETFs and the underlying basket. We consider normal trading periods as well as market events. Figure 2 shows the spread between the SPY weighted mid-price and underlying basket fair value (weighted mid-prices for individual stocks used in NAV formula, as described in Section 2.3) on July 15, 2024. The spread appears to be centered around \$0.00 which is what we expected. Spreads that do not equal \$0.00 indicate theoretical arbitrage opportunities. For our analysis, we consider price discrepancies that exceed a round-trip cost stack of approximately \$0.062 per share, comprising \$0.006 in exchange fees (\$0.003 per leg, two legs) plus 0.5 bps of slippage per leg applied to the ETF mid-price. Episodes that produce gross spreads below this threshold are unprofitable in expectation regardless of execution latency.

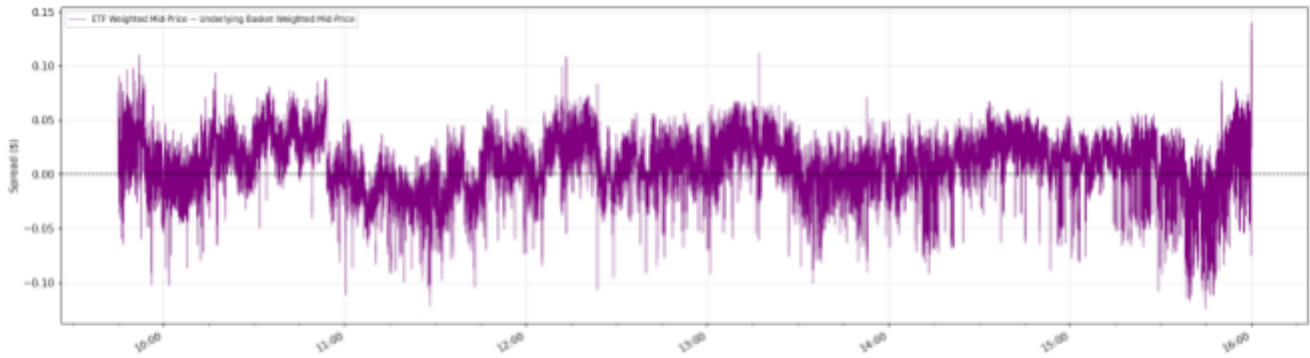


Figure 2. Spread between ETF (SPY) weighted mid-price and underlying basket fair value (weighted mid-prices for individual stocks used in NAV formula) on July 15, 2024.

3.1 July 2024

First, we consider arbitrage opportunities for SPY and VOO on trading days in the two week period between July 8, 2024 and July 19, 2024.

3.1.1 Data Re-Aggregation

One challenge we faced in analyzing the collected data for arbitrage opportunities was re-aggregating the data that was saved in 15 second periods, as discussed in Section 2.2. Our proposed solution to this challenge was to simply concatenate each 15 second period data file. This was feasible since the dataset only contained the bid, ask, and weighted mid-prices for the ETF and underlying basket.

3.1.2 Results (SPY)

In order to test the feasibility of the arbitrage strategy on SPY we measure the number of arbitrage opportunities and the speed and profitability of this strategy. We derive the following aggregated statistics shown in Table 1. We also consider the transaction costs involved in execution. Acting as a price-taker means buying at the ask price and selling at the bid price. Figure 3 shows the profit per singular share (i.e. one share of the ETF and one share of the underlying basket, corresponding to a proportionate share in each constituent) on July 15, 2024. Generally, because transaction costs are aggregated across ~500 constituent stocks, spreads are wider in the price of the underlying basket. This makes the arbitrage opportunity unfeasible in most cases, as executing each side of the trade at the bid or ask eliminates any opportunity. Thus, in Figure 3, the profit per share is below \$0.00 across the entire timeframe, approaching but not reliably crossing into positive territory

near market close. At the end of the trading day, volume spikes may briefly produce viable opportunities, though Table 1 shows zero net-positive trades on July 15 at any latency tier. In practice, we hypothesize the use of statistical arbitrage or creation and redemption could make this opportunity more feasible. See section 4.2 for a discussion of creation and redemption.

Date	Median Episode Duration (ms)	Mean Basis (bps)	Mean Gross \$/share	Total Net \$ (22 μ s)	Viable Opps (22 μ s)
2024-07-09	9.1	0.002	0.0048	0.000	0
2024-07-10	6.6	0.038	0.0085	0.104	6
2024-07-11	14.9	-0.048	0.0063	0.000	0
2024-07-15	7.6	0.173	0.0060	0.000	0
2024-07-16	1.4	0.259	0.0102	-0.006	2
2024-07-17	4.9	0.466	0.0057	0.042	2
Aggregate	7.4	0.148	0.0069	0.140	10

Table 1. Aggregated SPY arbitrage analysis for data from July 9, 2024 to July 17, 2024

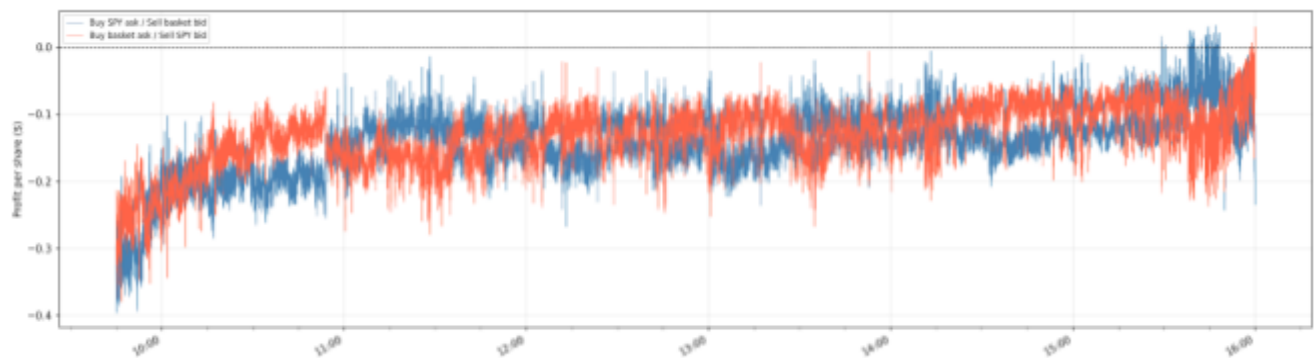


Figure 3. Profit per share of buying the ETF (SPY) at the ask price and selling the basket at the bid price or buying the basket at the ask price and selling the ETF (SPY) at the bid price July 15, 2024.

3.1.3 Results (VOO)

In order to test the feasibility of the arbitrage strategy on VOO, we similarly measure the number of arbitrage opportunities and the speed and profitability of the strategy. We derive the following aggregated statistics shown in Table 2. VOO produced 28 viable arbitrage opportunities at the optimistic latency tier across four July 2024 sessions, compared to 10 across six SPY sessions in the same period. This higher frequency reflects VOO's wider basis distribution, with standard deviations of 0.58-1.02 bps across sessions versus 0.49-0.71 bps for SPY. The aggregate mean basis on VOO was -0.072 bps versus 0.148 bps on SPY, and total simulated net P&L at the 22 μ s latency tier was \$0.568 per share versus \$0.140 for SPY. The pattern suggests that the arbitrage opportunity on the underlying basket is structurally more accessible on VOO than SPY during normal trading conditions, though both products remain unprofitable at most timestamps once transaction costs are accounted for.

Date	Median Episode Duration (ms)	Mean Basis (bps)	Mean Gross \$/share	Total Net \$ (22 μ s)	Viable Opps (22 μ s)
2024-07-09	14.0	0.096	0.0049	0.006	4
2024-07-15	9.4	-0.015	0.0068	0.376	16
2024-07-16	6.0	-0.226	0.0072	0.038	5
2024-07-17	12.0	-0.144	0.0083	0.149	3
Aggregate	10.7	-0.072	0.0068	0.568	28

Table 2. Aggregated VOO arbitrage analysis for data from July 9, 2024 to July 17, 2024

3.2 Leveraged ETF Case Study (SPXL)

We also explore arbitrage opportunities on SPXL, a leveraged S&P-500 ETF on trading days in the week of July 15, 2024. Figure 4 shows the ETF and basket prices over the trading day on July 15, 2024. We observe far more price volatility in the ETF due to use of leverage. It appears that arbitrage

opportunities are more plentiful. However, our fair value calculation may not be correctly taking into account the leverage used in the ETF, so we provide these results as a preliminary exploration. We calculate the fair value of the underlying basket by computing the NAV from the fair value of each underlying stock. We multiply the returns of the underlying basket by 3 to simulate the leverage in the ETF. However, this leads to the ETF consistently trading at a premium to its underlying basket, indicating that our calculation of the fair value of the underlying basket is incorrect. One trading strategy using the leveraged ETF could instead be to treat the price of the leveraged ETF as mean reverting around the unleveraged basket fair value; however, this may not be the case because of a concept called volatility decay, in which the price of the leveraged ETF falls even when the price of the underlying basket stays flat. We consider this concept beyond the scope of our current project.

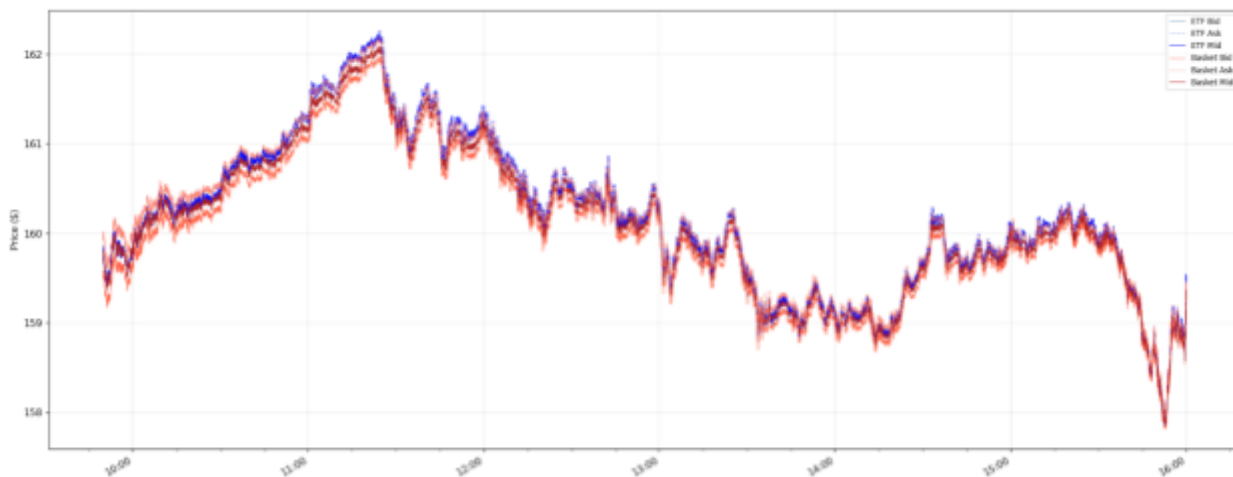


Figure 4. ETF (SPXL) and underlying basket bid, ask, and mid prices for July 15, 2024.

3.3 FOMC Meeting Case Study

During high market volatility and trading volume, arbitrage opportunities may become amplified, so we consider SPY during the Fed rate cut announcement in September 2024. Figure 5 shows the ETF and basket prices over the trading period. Evidently, at 14:00:00 the FOMC statement indicating the rate cuts was released. Table 3 shows aggregated results over the period before and after the rate cut announcement, indicating that arbitrage opportunities may have greater magnitude during market news events causing increased volume and volatility. Interestingly, opportunities appear for shorter periods during this event, perhaps indicating increased market efficiency.

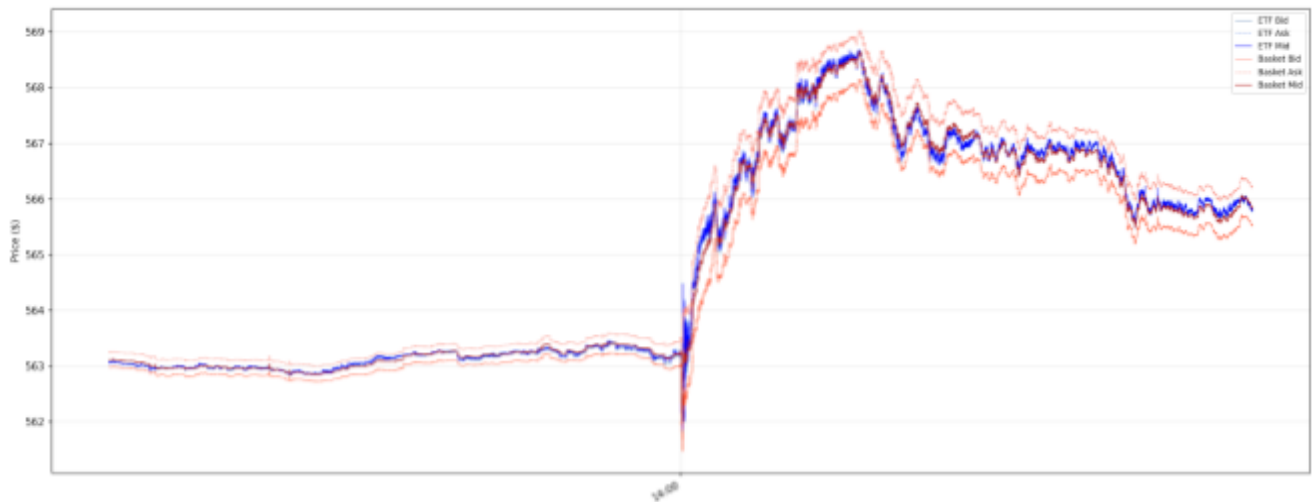


Figure 5. ETF (SPY) and underlying basket price before and after Fed rate cut announcement in September 2024.

Median dislocation time (ms)	Mean peak dislocation value (\$)	Maximum profit from period tested (\$)
0.1	0.1500	2.91

Table 3. Aggregated SPY arbitrage analysis for the period before and after the Fed rate cut announcement on September 18, 2024 (13:45–14:14 ET window)

3.4 Liberation Day Case Study

For similar purposes discussed in Section 3.2, we consider the trading day following President Donald Trump’s tariff announcements on April 2, 2025, referred to as Liberation Day. For this analysis, we consider the SPY on April 3, 2025. Figure 6 shows the ETF and basket prices over the trading period of April 3, 2025. Table 4 shows aggregated results over the period, again, indicating that arbitrage opportunities may persist longer and have greater magnitude during market news events causing increased volume and volatility. For reference, note the pure arbitrage opportunity between 12:00 and 13:00.



Figure 6. ETF (SPY) and underlying basket price during April 3, 2025 (day following Liberation Day).

Median dislocation time (ms)	Mean peak dislocation value (\$)	Maximum profit from period tested (\$)
1.7	0.0241	0.29

Table 4. Aggregated SPY arbitrage analysis for data from April 3, 2025 (day following Liberation Day)

3.5 ETF Pairs Trading Analysis

For this project, we also investigate a strategy that seeks to capture price discrepancies between S&P-500 ETFs themselves, rather than between the ETF and the underlying basket. This strategy is known as pairs trading. We consider 3 S&P-500 ETFs, SPY, VOO, and IVV. Figure 7 shows the return from open for each ETF in basis points (bps) on July 8, 2024. This value is calculated by dividing the current weighted mid-price and opening weighted mid-price for each ETF and represents a time scale of 1ms. 1 bp is equal to 0.0001. As illustrated in Figure 7, each ETF exhibits nearly identical returns, which is expected. Further, in Figure 8, we show the spread between each ETF return in basis points. In Table 5, we show the average spread between the returns of each ETF. Table 5 reports return-divergence statistics only for the SPY-VOO pair, since IVV data was available for the single July 8, 2024 session shown in Figures 7 and 8 but was not re-collected for the broader analysis windows used elsewhere in this report. These spreads exist, but are often less than 1 bp, making the opportunity seemingly unfeasible. However, we note that the steep price change before 15:00 causes

spreads to briefly spike. More analysis in this regard is required to explore this pairs trading strategy on trading days with more market price volatility.

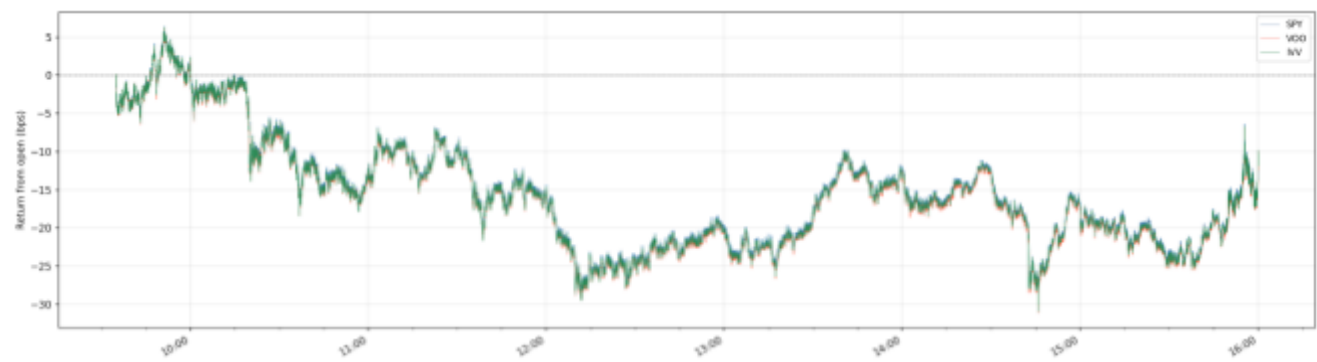


Figure 7. Returns in basis points for SPY, VOO, and IVV on July 8, 2024.

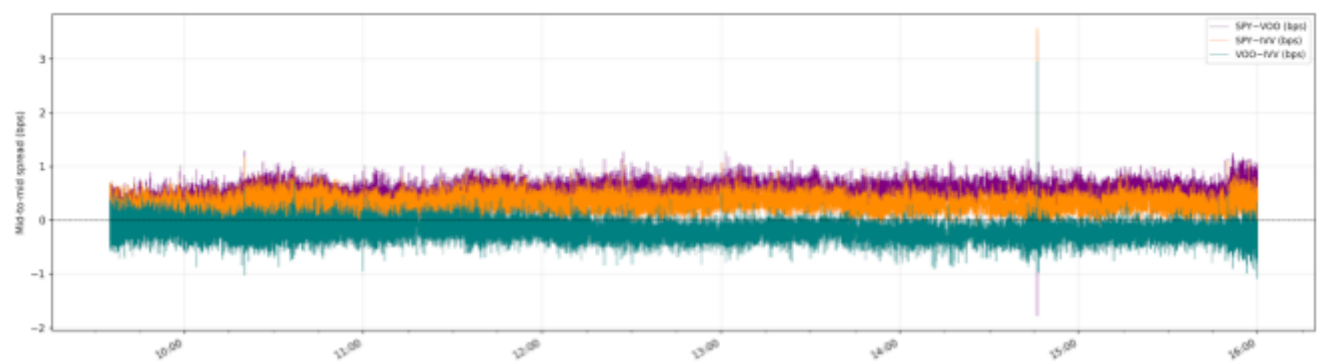


Figure 8. Spread between returns for each S&P-500 ETF pair considered.

Date	Mean Return-Di vergence (bps)	Std (bps)	Peak-corr Lag (ms)	Peak Corr
2024-07-09	0.239	0.142	0.0	0.395
2024-07-15	0.050	0.152	0.0	0.476
2024-07-16	0.203	0.152	0.0	0.441
2024-07-17	0.204	0.128	0.0	0.456
Aggregate	0.174	0.144	0.0	0.442

Table 5. Average return spread between SPY and VOO across four July 2024 sessions where both ETFs had data coverage

3.6 Lead-Lag Relationship Analysis

Across all six SPY July sessions and the FOMC and Liberation Day windows, the cross-correlation between ETF and basket returns peaks at lag zero in 100% of rolling 10-minute windows on a 10 ms grid. We do not observe a consistent directional lead-lag relationship at this resolution. Any ETF-basket lead-lag effects, if they exist, occur on a sub-millisecond timescale below the resolution of our analysis.

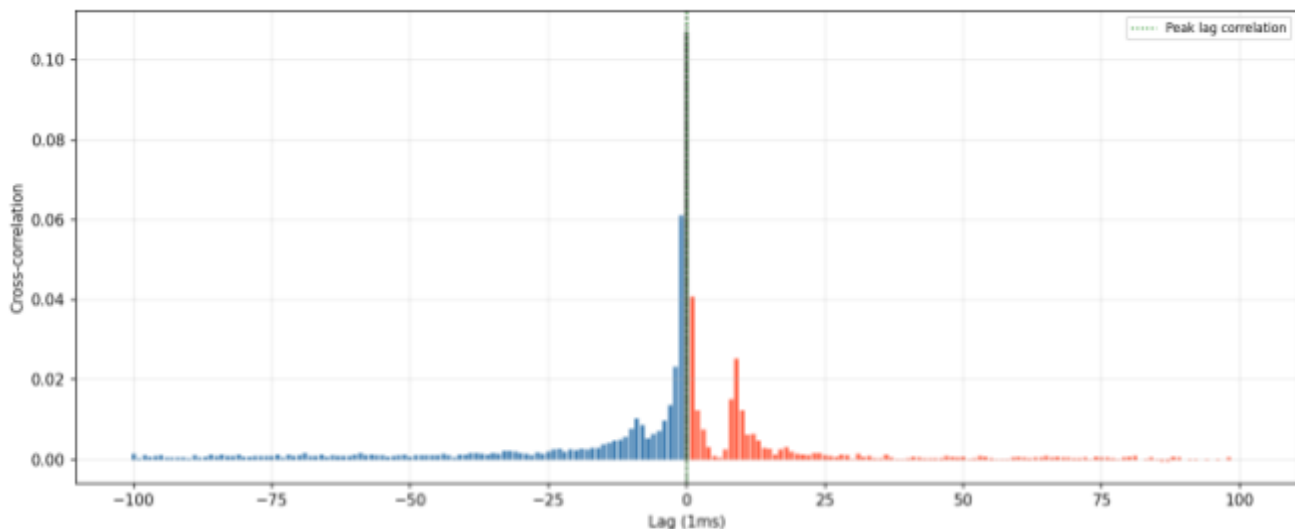


Figure 9. Cross-correlation between ETF returns and leading or lagging basket returns on July 15, 2024. Negative lag indicates that the basket leads the ETF, and positive lag indicates that the ETF leads the basket.

4. Discussion

4.1 Trading Strategy

We consider the following an example of what a trading strategy would look like for S&P-500 ETF arbitrage. When the NAV, which we consider the underlying basket fair value price, and the market price of an ETF differ by at least a given threshold, in other words when the spread reaches a certain value, execute the arbitrage. If NAV is greater than market price, buy the ETF and short the underlying basket. If NAV is less than market price, short the ETF and buy the underlying basket. All ~501 transactions must be as close to simultaneous as possible.

4.2 Why Authorized Participants Exist

Special entities called authorized participants (APs) serve as the link between the stock market and ETF providers. In a process called creation, the AP gives a basket of stocks to the ETF provider and gets an ETF share back. This is used to correct a premium, where the ETF is greater than the NAV. In a process called redemption, the AP gives ETF shares back to the provider and gets the basket of stocks in return. This is used to correct a discount. These transactions all occur at NAV instead of market value. This incentive helps keep spreads small, making the market extremely efficient. Authorized participants are able to more effectively capture arbitrage spreads without paying transaction fees on both sides of the trade. However, authorized participants are typically required to exchange large quantities of securities. For example, for creation in SPY, APs must assemble creation

units consisting of the underlying securities of the fund. These creation units must typically be equivalent to 50,000 ETF shares (State Street Global Advisors, n.d.).

4.3 How Fast Do You Need to Be

An optimized C++ engine can execute trades in 10-30 μ s. Sequentially processing the 500 underlying stocks adds more latency, and risks stock prices being stale by the time the loop is done. On an FPGA, parallel logic gates allow for concurrent computation of all 500 underlying stocks in a single clock cycle, allowing a much faster execution time of 0.5 μ s. Our findings are that arbitrage opportunities last for longer, meaning that both software and FPGAs are able to take advantage with correct strategies. However, this edge is not guaranteed, and we hypothesize based on our findings that these markets are extremely efficient, resulting from many competitors competing for these fleeting opportunities.

4.4 Conclusions

Opportunities for ETF arbitrage exist but require robust data analysis to compute fair value accurately and extremely fast execution. Our analysis did not reveal any episode that simultaneously satisfied a basis magnitude greater than 5 bps and a duration greater than 100 ms across any of the eleven sessions analyzed, the rough threshold at which an authorized participant creation/redemption cycle would become profitable. Even during the FOMC announcement, dislocations exceeding 5 bps consistently resolved within a millisecond. Further work will be necessary to decipher imperfections in our fair value estimates, and real, longer-term, price discrepancies. Overall, the profit margins appear to be very slim because the market is efficient. Additionally, transaction costs and latency are difficult to remedy.

4.5 Future Work

For future applications, more data should be gathered to make our analysis more robust. Data collection, as discussed in Section 2, presented our greatest challenge. Additionally, comparison with published intraday net asset value (iNAV), which is typically published every 15 seconds, would be helpful in validating our methods. However, we found iNAV data was difficult to obtain for the scope of this project. Most of our analysis also took place after the market opened, since it was difficult to measure the fair value from opening prices because some quotes were erratic during this period. For future work, it would be interesting to implement a practiced method for estimating opening prices.

Finally, the leveraged-ETF fair-value model used in Section 3.2 applies a constant 3x return multiplier and does not account for the daily-reset mechanic, expense-ratio drag, or intraday rebalancing characteristic of products like SPXL. A more rigorous fair-value derivation for leveraged ETFs would tighten the conclusions of that case study.

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